NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

"Explorations and treatments by the osteopath who is worthy of the title D.O. should deliver 999 children out of 1000 with no instrument but his hands; otherwise, he is not fit to be trusted with birth of child and care of mother and offspring..."

-Andrew Taylor Still:

<u>Philosophy and Mechanical Principles of Osteopathy</u> p 210



ORIGINAL CLASS IN OSTEOPATHY, 1892-93
(Photo courtesy Still National Osteopathic Museum, Kirksville)

- To review pertinent anatomy and how it is central in understanding treatment of the whole person (autonomic nervous system, circulation, lymphatics, Chapman's points Gyn Lect)
- To highlight structural and physiologic changes of pregnancy and how the osteopath can be of service
- To discuss Ante, Intra and Post-Partum Considerations of Common Pregnancy Complaints
- Highlight Osteopathic Research in the field and How YOU can and will Help. i.e. recognizing and treating somatic dysfunctions/ facilitation related to nerve, artery, vein, lymph, organ etc.
- Know contraindications to manipulation

Source: Course Syllabus

Center of Gravity Change

NEW YORK INSTITUTE OF TECHNOLOGY

↑ Lumbar and Cervical Lordoses

↑ Thoracic kyphosis OR Flattening

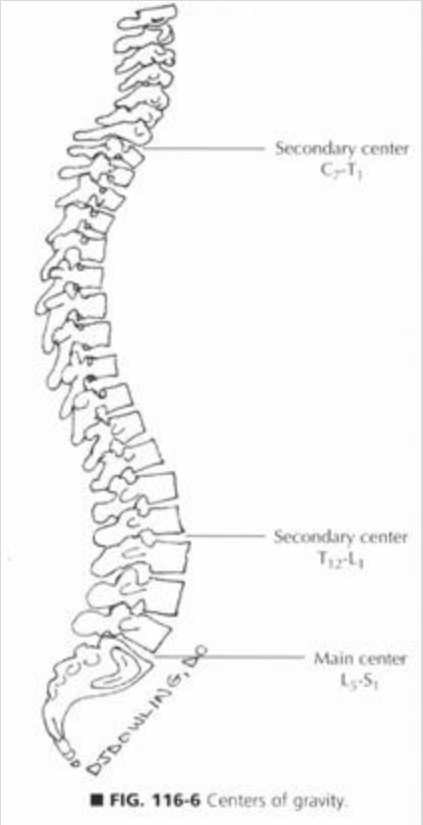
↑ Lumbar disc stress

↑ Anterior pelvic tilt

Sacral dysfunctions

"Sacral sag – Fascial drag"

Source: DiGiovanna



■ FIG. 116-6 Centers of gravity.

NYIT College of Osteopathic Medicine

Structural

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Forward tilt of head

Junctional Strains

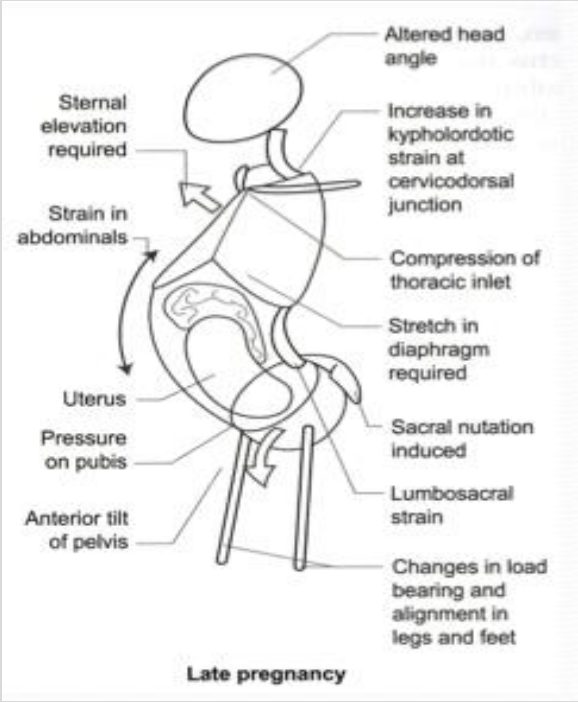
Diaphragm Restrictions

Rectus abdominus Weakening

↑ Pubic pressure & laxity

Anterior pelvic tilt

Change in ankle & foot alignment /Gait



Center of Gravity Shifts

Anterior Angulation of Cervical Region

↑ Thoracic kyphosis OR ↑ flattening

↑ Lumbar Lordosis

↑ Sacral Nutation

Knee Hyperextension

Gravitational Fascial strains

NYIT College of Osteopathic Medicine

Source: Kendi Hensel, DO

Thoraco-Lumbar (T-L) Junction

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Area of Musculoskeletal Strain in pregnancy
- **Good motion important for proper Diaphragmatic excursion**
  - Crus of posterior diaphragm attachment
- **T10-L2** SNS for **bowel, uterus and adnexa, bladder and ureters**, adrenals and renal
  - any facilitation at these somatic segments by one organ can affect other organs supplied by the same segments
- *Exacerbation of facilitation of this area is common after C-Section*

Cardiovascular

Increased plasma volume

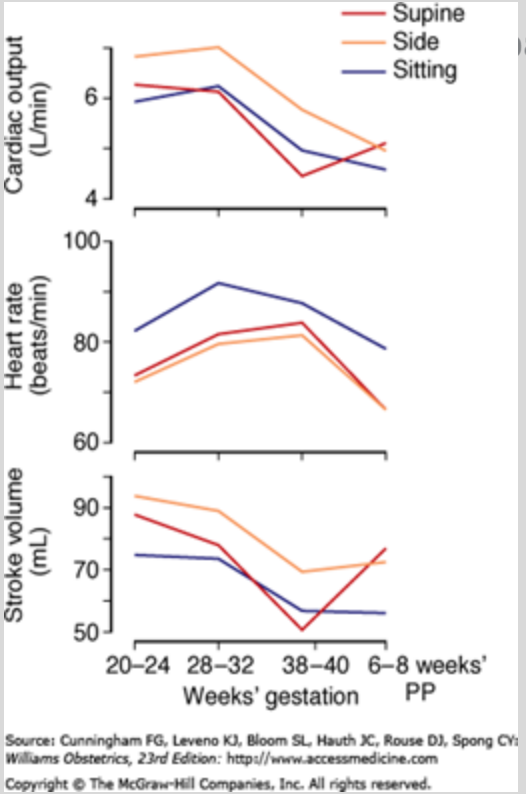
Decreased systemic vascular resistance

- Progesterone vasodilation

Enlarging uterus compresses IVC

- ⍰ Decreased venous return

Left lateral recumbent position alleviates



Source: Cunningham FG, Leveno KJ, Bloom SL, Hauth JC, Rouse DJ, Spong CY: *Williams Obstetrics, 23rd Edition*: http://www.accessmedicine.com

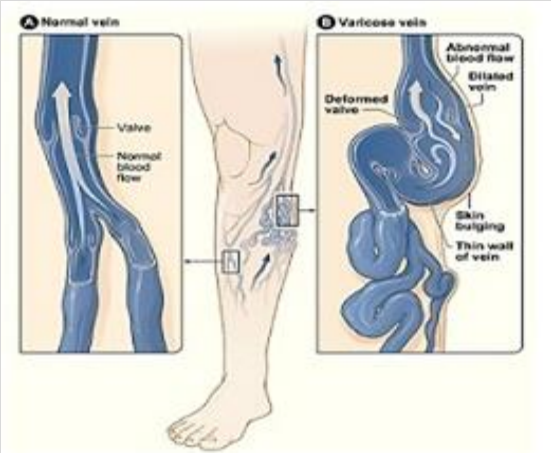
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Hematology

Hypercoagulable state
- Estrogen stimulates clotting factor synthesis
- Venous stasis → DVT

Increased plasma volume, venous stasis, increased vascular hydrostatic pressure
- Varicosities
- Hemorrhoids
- Organomegaly (congestion)
- Carpal Tunnel Syndrome
- Pedal Edema



NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

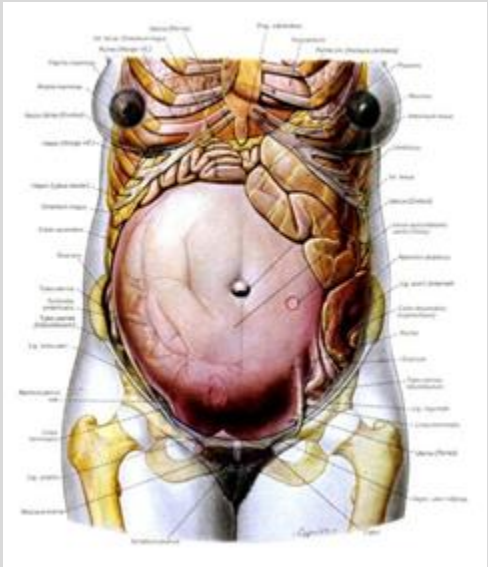
Respiratory

Diaphragm displaced 4 cm

Displaced diaphragm and tight crus may hinder lymphatic drainage

Decreased lymph flow organomegaly/edema





Source: Netter

NYIT College of Osteopathic Medicine

GI

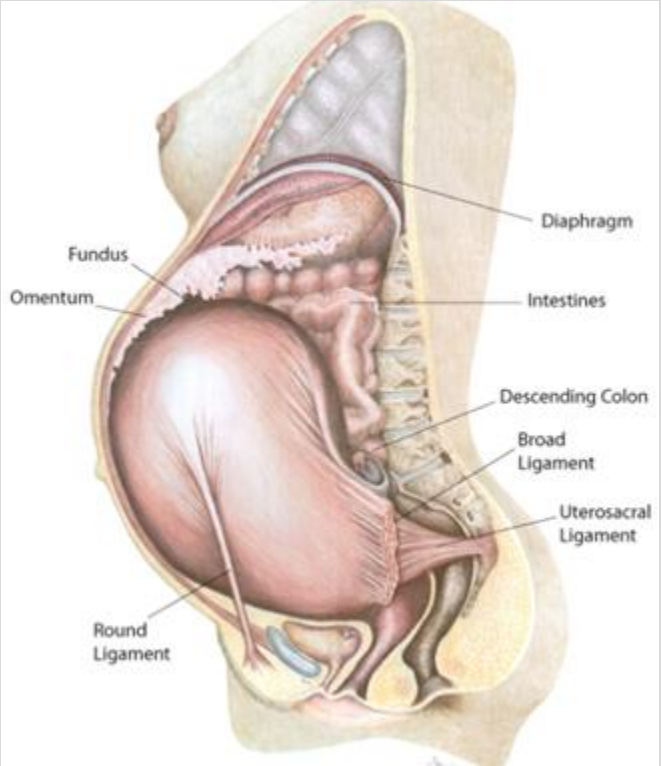
NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

↑ Cholesterol gallstones

“Morning” sickness

Increased estrogen

Hyperemesis gravidarum



NYIT College of Osteopathic Medicine

GU

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

↑ Frequency and urgency
- Compression of bladder
  - Stress incontinence

Enlarged uterus impairs lymph drainage and bladder emptying
- ↑ UTI – urinary tract infections



Endocrine

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Pituitary enlarges

- Increased levels of prolactin & oxytocin
  - *PRO*lactin → milk *PRO*duction
  - Oxytocin → milk let down and uterine contractions

Thyroid

- Increase in Total T3 &T4 and TBG;
- Normal levels FREE T3,4 -remain euthyroid

Physiologic hyperparathyroidism

- Increased calcium for baby

Endocrine Hormone: Relaxin

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Derived mostly from corpus luteum
  - Found softening of pubic symphysis
  - Greater distensibility/ pliability
    - Accommodate changes of pregnancy

- Pregnant women: Clinically “easier to treat with OMT”

Serum levels elevated throughout pregnancy until PPD (post-partum day) #3-4

- Possibly remain in the *tissues* up to 6-12 weeks after delivery

- When does Relaxin reach it’s peak??

“Normal” Labor Times

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

	Nulliparous	Multiparous
1st Stage- Latent Phase	< 20 hrs	< 14 hrs
1st Stage- Active Phase	1.2 cm/hr ~ 8hrs	1.5 cm/ hr ~ 7hrs
2nd Stage	< 2 hrs (3 w/ epidural)	< 1 hr (2 w/ epidural)
3rd Stage	~ 10 mins	~10 mins
TOTAL TIME	<30 hrs	< 22 hrs

Labor and Birth

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Five "**P** s" of Labor

Passageway – maternal pelvis and soft tissues

Passenger – baby

Powers:

1°: involuntary uterine contractions
2°: voluntary abdominal mu. and diaphragm

Position

Psyche

Medications and Pregnancy

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Increased Absorption
- Slowed motility GI
- Delay onset controlled-release

OR No absorption
- Hyperemesis/ Reflux

Fluid/Adipose change
- Altered dynamics for water & fat soluble medications

Placenta cytachrome P450: metabolizes drugs

Despite Major Advances

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Pharmaceuticals are generally avoided due to potentially harmful effects to fetus

OMT has been used:

- Pain complaints (LBP, HA, labor, post-op…)
- Edema
- GI complaints: constipation, GERD, emesis
- Reduce labor time
- Post partum depression

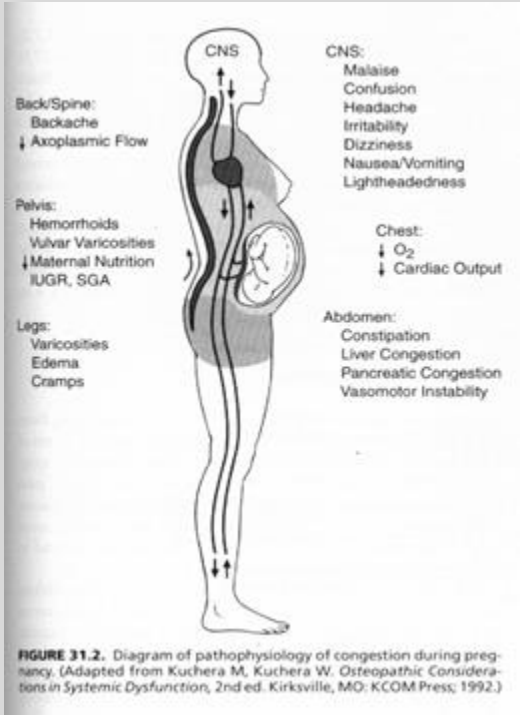


FIGURE 31.2. Diagram of pathophysiology of congestion during pregnancy. (Adapted from Kuchera M, Kuchera W. *Osteopathic Considerations in Systemic Dysfunction*, 2nd ed. Kirksville, MO: KCOM Press; 1992.)

NYIT College of Osteopathic Medicine

Historical Reports

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

1918 Study:

LABOR TIME on average:

- Primiparas who had OMT: 9 hrs 20 min
- Primiparas had NO OMT: 15 hrs

- Multiparas who had OMT: 5 hr labor time
- Multiparas had NO OMT: 9 hrs

1911 Study:

Primiparas ***with OMT*** during pregnancy avg labor **9h 54 m**.

Multiparas ***with OMT*** average labor: **6 h 19**

Primiparas ***NO OMT*** averaged **21h 6m** labors.

Multiparas ***NO OMT*** averaged ***11h 41m***

JAOA 2020: Effect of Osteopathic Obstetrical Management on the Duration of Labor in the Inpatient Setting Martingano et al.

Results: 100 patients enrolled. Fifty patients underwent adjunctive OMT in addition to standard labor management were matched to controls who received std labor management only. The mean (SD) **Labor duration** for patients receiving **OMT was significantly shorter** (11.34 [6.62] hours [range, 1.1-27.0 hours] vs. the **labor duration for controls** 16.57 hours [4.39] [range, 1.0-58.8 hours], respectively; **P=.03**).

NYIT College of Osteopathic Medicine

Research
Paraspinal Inhibition

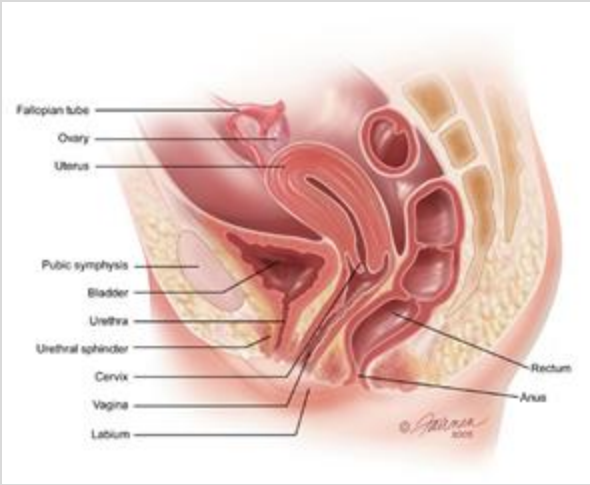
NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

500 patients
70.4% lumbar LBP (low back pain)
2 groups

1) Received **steady lumbar pressure "paraspinal inhibition"** at L1-5 during contractions

vs.

2) No treatment



NYIT College of Osteopathic Medicine

NIH Study Treatment Protocol

- Sitting
  - Articulatory T-spine
- Supine
  - Cervical ST/MFR
  - Thoracic Inlet/MFR
- Lateral Recumbent (R and L)
  - Scapulothoracic MFR
  - Lumbosacral ST

- Supine
  - Ab Diaphragm
  - Pelvis
    - AP Pelvic diaphragm Unwinding
    - SI articulation
    - Frogleg
    - Ilial Rotations
    - Pubic decompression
  - CV4

NYIT College of Osteopathic Medicine

Source: American Journal or Obstetrics “Osteopathic manipulative treatment of back pain and related symptoms during pregnancy: a randomized controlled trial" John C. Licciardone, DO, MS, MBA et al.

Osteopathic
Diagnosis & Treatment
What makes you
even more special:

Your ability to RECOGNIZE,
DIAGNOSE & TREAT “Somatic dysfunctions” and to INTEGRATE this ability into the care of your patients.

NYIT College of Osteopathic Medicine

OMT

Treat what you find.

Be sure to check:



Transition zones/ junctions

OA, C-T, T10-L2, L-S, SI's

Sacral mobilization/ rocking

Lymphatic drainage

Axillary folds, abdominal diaphragm,
T.O., mesenteric lifts…

Cranial

SBS, C2, OM Suture …

NYIT College of Osteopathic Medicine

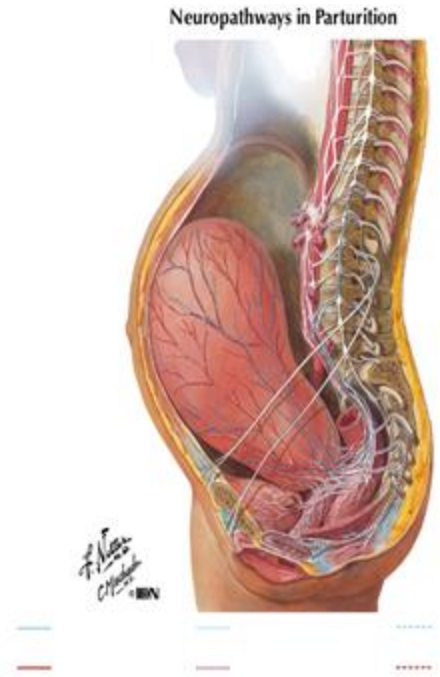
Transition Zones and Sacrum

T-L junction: may help regulate contractions

- **Paraspinal inhibition*
- **Seated articulatory technique*
  - SNS to uterus T12-L2

Sacrum may help regulate cervical dilation

- *SIJ articulation* via PSNS to cervix S2-4



Wrist dysfunction



2nd most common Musculoskeletal c/o pregnancy

As many as 35%

CTS (carpal tunnel syndrome), DeQuervain's tenosynovitis, Trigger finger

Fluid retention/Hormones

More common in pts with HTN/ pre-eclampsia

Ante-partum: typically 2nd,3rd Trimesters

Post-partum with lactation: Tail of breast gets engorged in axilla – may inhibit lymph drainage from UE.

Somatic Dysfunction (SNS to **Upper Extremity T1-4 or 8**)

Lymphatic Drainage



Lymphatic movement

- *Diaphragms*
- *Abdominal lifts*

Carpal Tunnel Release

Scapular release

Help swollen breasts and tenderness
Encourage milk let down

Mastitis

- *T1-8 (especially T4-6) Somatic dysfunction*
- *Anterior/posterior ribs*
- *Thoracic Outlet*

NYIT College of Osteopathic Medicine

Cranial

SBS

Encourage movement of CSF

Decrease “morning sickness”/ hyperemesis gravidarum

*OM Suture □ Vagus

(Thoracic Diaphragm □ Liver decongestion)

Treatment of cervical spine

C2 –

Dura attachment

Vagal considerations

CV4

Treatment of the Cranium has been shown to influence uterine contractions.

Gitlin RS, Wolf DL:Uterine Contractions follov
Osteopathic Cranial Manipulation – a pilot stu
JAOA 1992;92:1183

May help stalled labor

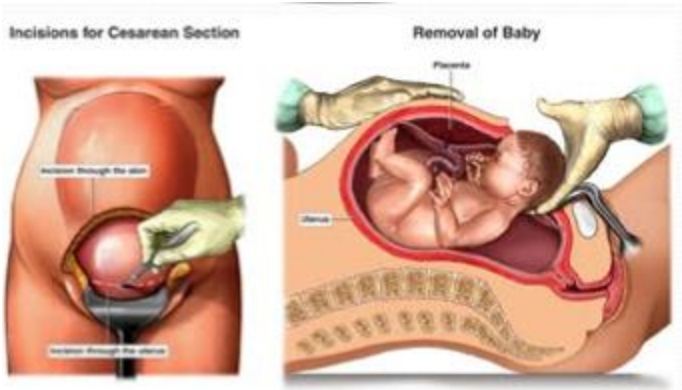


C-Section

Types
- Low Transverse (*MC)
  Dermatome ~T12
- Vertical
  Dermatome T7-T12

Surgical Incision -->
Acute Dermato-somatic reflex
-->
Segmental Facilitation -->
Increased Pain

Treat areas of facilitation
With gentle OMT:
PARASPINAL INHIBITION



Immediately Post-Partum

Assist SI joint and spine return to “normal” after pelvic distortion made possible by maternal hormones & legs up in stirrups/ lithotomy position:

1. FLEX (hips & knees over abdomen)
2. ADDUCT thighs
3. INTERNALLY ROTATE hips
4. EXTEND legs carefully onto bed

<u>In first few days to weeks</u>:

Try to address Junctional strains (mu, fascia, ligaments), Autonomics, Cranial

Post-Partum Care

Assist pelvis, axial skeleton, and supporting soft tissue to functional non-gravid state w/ osteopathy

Evaluate the sacrum as delivery encourages an anterior sacral base and extension stain pattern of the cranio-sacral mechanism which can lead to fatigue, depression and low energy

"Sacral-sag and fascial-drag lead to …."

And last but not least ☺

Summary Slide



- For the curious physician, many pearls of wisdom can come from caring for this special population
- Always strive to make sense of anatomic variation and how it impacts physiology
- Satisfaction comes from continual learning and from helping others – osteopathy is great for that.

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

References:

- **REQUIRED READING:**
- **1) AT Still Biography pg 202-206**
- **2)** An Osteopathic Approach to Diagnosis and Treatment: DiGiovanna, Chp 116 3rd and 4th edition

- **REFERENCES:**

- JAOAJune 2019 Effect of Osteopathic Obstetrical Management on the Duration of Labor in the Inpatient Setting: A Prospective Study and Literature Review Daniel Martingano, DO, George C. Aglialoro, DO et al
- AJOG articles: 1)"**Osteopathic manipulative treatment of back pain and related symptoms during pregnancy: a randomized controlled trial" John C. Licciardone, DO, MS, MBA et al. 2) PROMOTE study**
- Williams Obstetrics
- Foundations of Osteopathic Medicine 2nd Ed p 450-461
- Osteopathic Considerations in Systemic Dysfunction (Obstetric patient p 149-58)
- Whiting LM, Can the length of labor be shortened by osteopathic treatment? JAOA 1911;11:917-21
- Hart LM. Obstetrical Practice. JAOA 1918;609-14.
- King et al., Osteopathic Manipulative Treatment in Prenatal Care: A retrospective study - JAOA, Vol 103, No 12, Dec 2003
- Gitlin RS, Wolf DL Uterine Contractions following Osteopathic Cranial Manipulation – a pilot study JAOA 1992;92:1183

Question

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

A 30 year old G2P2 female presents to your office complaining of low back pain and constipation. 6 weeks prior she had a healthy baby girl via low transverse C-section due to 'failure to progress' during labor. Osteopathic structural exam will most likely reveal tissue texture changes at:

- A- T10-T11
- B – T5-9
- C – T12 – L2
- D – T1-4
- E – C4-6

Question

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

A 32 G3P2 at 15 weeks GA presents to your office complaining of headaches and neck strain. She states she would like to take something for the pain, but fears it may affect the child. She asks what are the potential issues in regard to taking pain medications during pregnancy. You reply:

A – There are no adequate well controlled human studies demonstrate no risk to the fetus in the first trimester

B – There may be altered absorption of the pain medication

C – There may be altered distribution of the medication

D – There may be altered metabolism of the pain medication

E – All of the Above

Thoracic Inlet Myofascial Release

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine





Lateral Recumbent Scalpulothoracic MFR

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine





Lateral Recumbent Lumbosacral Soft Tissue

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine



Supine Diaphragm Release

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine





Frogleg
Sacral Articulation

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Contact is on sacrum with fingers at the base and palm at apex
- Pt's hips and knees are flexed with feet together and knees falling to side
- Sacrum is taken to point of ligamentous balance with respiratory assistance
- As patient holds breath in most useful phase, she straightens out legs to rotate innominates
- Recheck

Kimberly p194

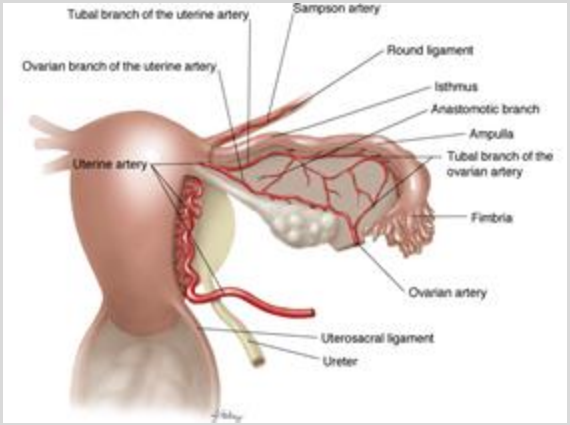
Arterial Supply: Ovarian artery

from aorta ~level of L2, Anastomose with branches of uterine artery.

Venous Drainage:

Right: ovarian vein to the IVC

Left: to renal vein



Lymph Drainage: peri-aortic

PSNS: Vagus

SNS: T10-12

"*Mittelschmerz" referred through afferents along T10 dermatome – periumbilical area*

NOTE: Ovary often close to appendix